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January 26th, 2026

ITAI 2376 - Deep Learning

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A01 A Comparative Analysis of Machine Learning and Deep Learning Tools and Frameworks - NLTK vs spaCy

Two widely used Python NLP libraries include spaCy and NLTK. spaCy was publicly introduced in February 2015 by Matthew Honnibal with the hopes of creating a fast, production oriented NLP for real applications. A key advantage of spaCy is its high speed capabilities due to it being built as a professional power tool. It works with pre-trained models that perform tasks such as tokenization, part-of-speech tagging and NER. It also has the capability to process text all at once. This makes spaCy well suited for applications that require scalability and deployment in production environments (like chatbots and text classification tools).

NLTK (Natural Language Toolkit) was originally developed in 2001 at the University of Pennsylvania by Steven Bird and Edward Loper. NLTK is great for learning NLP techniques such as stemming, parsing, and semantic analysis. A strong feature of NLTK is its built-in access to numerous linguistic datasets and corpora (like WordNet) that are commonly used in academic research and classroom settings.

Even though NLTK is powerful for experimentation and understanding how NLP algorithms works, it is slower and requires a more manual setup compared to spaCy. This means NLTK is highly valuable for education, research, and prototyping while spaCy works best in performance and scalability. Both of these libraries serve important roles for Natural Language Processing.

Sources:

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